MA QI

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Work Experience

Zhejiang University Cognomics Lab | Research Assistant | PI: Kong, X. Z. Jul 2025–Present

Conducting interdisciplinary research at the intersection of cognitive neuroscience and artificial intelligence, with a focus on brain hemispheric lateralization.

Education

Zhejiang University | Bachelor of Science in Psychology | Hangzhou, Zhejiang Sep 2021–Jul 2025

Completed systematic training in psychology and statistical methods, with further coursework in programming, data analysis, and machine learning.

Research Experience

Reconstruction-Derived Specificity in Schizophrenia Jun 2026–Present

Cognomics Lab
PI: Kong, X. Z.

- **Research Overview:** Analyze structural MRI data from 948 participants across five sites to investigate cross-hemisphere reconstruction-derived neuroanatomical specificity (ANS/RNS) in schizophrenia.
- **Research Highlights:**
  - **Multiscale group comparisons:** Test whether ANS/RNS differ between patients with schizophrenia and healthy controls across spatial scales.
  - **Clinical and cognitive associations:** Examine relationships between ANS/RNS and clinical and cognitive measures to assess the phenotypic relevance of reconstruction-derived measures.
  - **Disease classification:** Evaluate whether ANS/RNS features can distinguish patients with schizophrenia from healthy controls.
- **Manuscript:** In preparation.

Reconstruction-Derived Specificity and Handedness Oct 2024–Present

Cognomics Lab
PI: Kong, X. Z.

- **Research Overview:** Used handedness as a behavioral lateralization phenotype to test whether cross-hemisphere reconstruction residuals capture handedness-related structural information beyond conventional gray-matter volume asymmetry indices.
- **Research Highlights:**
  - **Cross-hemisphere reconstruction:** Used a deep generative network (DGN) to reconstruct one hemisphere from the other. Expressed residuals between original and reconstructed images as absolute neuroanatomical specificity (ANS) and relative neuroanatomical specificity (RNS).
  - **Global specificity and validation:** Moved beyond homologous-region comparisons by using reconstruction residuals to capture whole-hemisphere structural relationships. Evaluated anatomical plausibility, individual specificity, test–retest reliability, and hemisphere-identity classification.
  - **Handedness analysis:** Compared ANS and RNS with a conventional gray-matter volume asymmetry baseline in the balanced AOMIC Hand190 cohort. Mean RNS AUC was 0.624, significantly higher than the baseline value of 0.539 ($p = 0.018$).
- **Manuscript Draft:** Led preparation of the English manuscript draft, *Reconstruction-derived hemispheric specificity captures distributed structural information related to handedness*.

Project Experience

HemiSpec: Hemispheric Specificity Toolkit | HemiSpec May 2026–Present

- **Project Overview:** Packaged an existing cross-hemisphere DGN and ANS/RNS framework into installable software. Researchers can generate bilateral measures from standardized gray-matter maps, conduct ROI and validation analyses, and integrate the measures into their own neuroimaging studies.
- **Technical Highlights:**
  - **Workflow integration:** Integrated FSL gray-matter preprocessing, bidirectional DGN reconstruction, ANS/RNS calculation, ROI export, and hemisphere-classification validation in one workflow.
  - **Reusable interfaces:** Built Python API, CLI, and GUI interfaces for batch generation of measure maps, ROI tables, and optional validation outputs.
  - **Software delivery:** Provided source-install instructions, bilingual usage guides, and pytest regression tests; documented input requirements, model dependencies, and outputs.

Laboratory Computing Cluster Deployment & Operations Jun 2026–Oct 2026

- **Identity migration:** Migrated 23 users and 5 POSIX groups from NIS to FreeIPA, preserving UID/GID, home paths, and login shells; resolved ID-range/SID mapping issues and verified Kerberos authentication.
- **Client migration:** Integrated 3 Linux clients with FreeIPA/SSSD and retired their NIS authentication, using prechecks, backups, staged cutovers, and rollback preparation; verified identity resolution, shared-home access, and directory permissions.
- **Compute scheduling:** Contributed to deployment and validation of a 4-node Slurm cluster; tested CPU batch jobs, RTX 3090 GPU allocation, interactive jobs, cancellation, and input/output file transfers as a regular user.
- **Operational handover:** Established account-provisioning, group-permission, and temporary-credential delivery procedures; authored FreeIPA first-login guidance and Slurm user and administrator manuals.

Cortical & Subcortical Visualization Codex Skills | Cortex & Subcortex May 2026–Aug 2026

- **Project Overview:** Packaged cortical and subcortical ROI-table visualization into Codex Skills, enabling natural-language interaction through an AI agent and reducing repeated tutorial lookups and ad hoc scripting.
- **Technical Highlights:**
  - **AI-agent interaction:** Built reusable workflows for natural-language-assisted atlas selection, figure generation, and result presentation.
  - **Cortical workflow:** Built on ggseg/ggseg3d to support multiple cortical atlases.
  - **Subcortical workflow:** Integrated Annie Bryant's *subcortex_visualization* for 2D maps of subcortical structures (including the thalamus), the cerebellum, and the brainstem.

Building a Language Model from Scratch | LoreWormGu Dec 2024–Mar 2025

- **Project Overview:** Built a language model with tens of millions of parameters from scratch, completing a training pipeline spanning pretraining, supervised fine-tuning (SFT), and R1 distillation fine-tuning.
- **Technical Highlights:**
  - **Architecture and tokenizer:** Implemented a Llama 3-style model with RMSNorm, configurable key/value heads, SwiGLU, and RoPE, alongside a BBPE tokenizer for Chinese text.
  - **Pretraining and optimization:** Trained on the Jiangshu dataset with custom token-level loss, forward passes, gradient accumulation, and mixed-precision training.
  - **SFT and R1 distillation:** Prepared open-source instruction data for supervised fine-tuning and used DeepSeek-R1 responses for response-based distillation.

N-back Working-Memory Load Classification | DecodeWM Jun 2023–Sep 2023

- **Course project:** Used HCP task-fMRI ROI features with logistic regression and random forests to distinguish 0-back from 2-back conditions in the Neuromatch Academy computational neuroscience course.

Science Communication

A Man's Brainhole | WeChat Official Account Jun 2026–Present

- **Editorial work:** Write Chinese-language explainers on brain science, computational neuroscience, NeuroAI, and research tools by selecting papers and explaining their methods and findings.
- **Production and quality control:** Use AI-assisted workflows for literature review, figure curation, typesetting, and publishing; verify sources, terminology, and quantitative claims and communicate limitations.
- **Audience growth:** Published **95 articles** and gained **1,200+ new followers**.

Skills & Languages

Skills: C, Python, machine learning and deep learning, probability and statistics, structural neuroimaging processing and visualization, Linux administration, FreeIPA/SSSD, Slurm.
TOEFL iBT: 5/6 (100/120).